

Workshop warm up tasks

The following tasks are aimed to give you a sense of the **data types and methods** used in the workshop. It will not take you more than 5-10 minutes to think about these questions. They will help us start off the use cases of the workshop.

Text analysis

Suppose that you have a bot that receives the following text as input:

```
----- >From: nick@gmail.com [mailto: nick@gmail.com] >Sent: 16 October 2022 17:42  
----->To: 'james@gmail.com >Subject: machine learning workshop -----> Hi  
James, how are you doing? </>Are you attending the workshop? <> The title of the workshop is  
catchy. It is offered by EPFL, here is their website https://www.extensionschool.ch> By the way,  
are you familiar with machine learning?...
```

And gives back to you the following **cleaned text**:

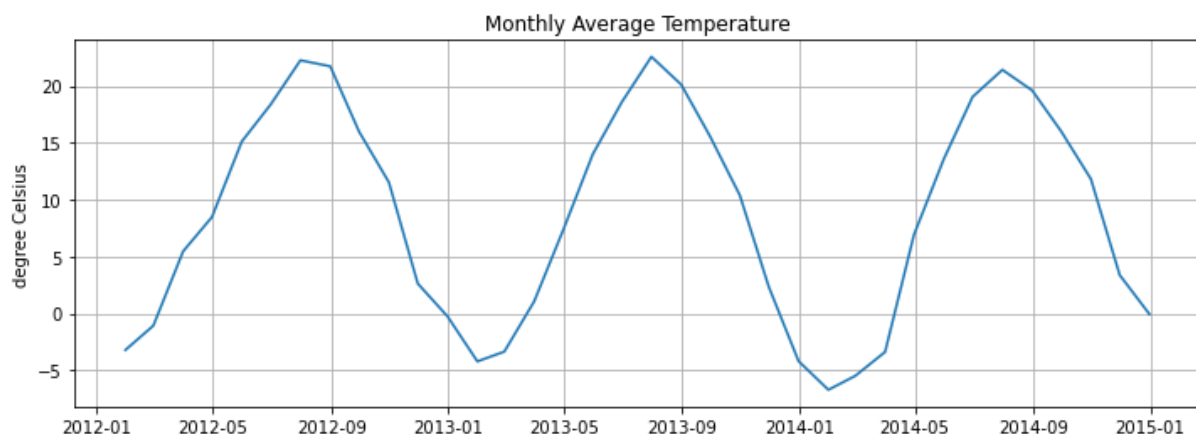
october subject machine_learning workshop attend workshop workshop catch offer epfl website
familiar machine_learning

Q1. What are the most frequent words in the cleaned text?

Q2. Can you guess the topic of the text based on the most frequent words?

Time series analysis

The following time series shows the monthly temperature until **December 2014**



Q1. What is the most evident characteristic of this data?

Q2. What would be a reasonable approximate temperature for January 2015?

Anomaly detection

Imagine you are observing the usual morning traffic on the way to work.

Q: What makes a car stand out for you?